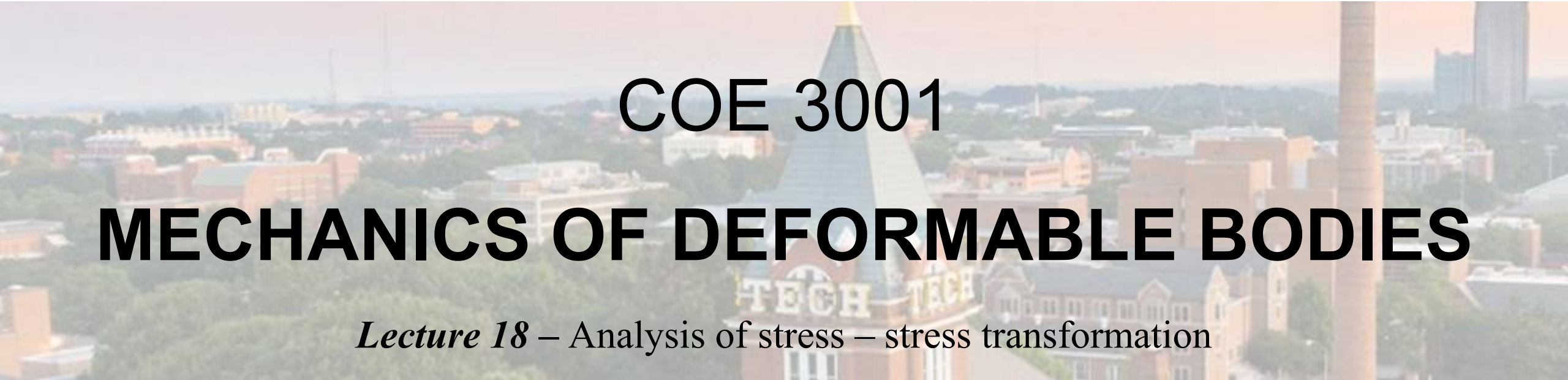




Georgia Tech
College of
Engineering

A wide-angle, slightly hazy photograph of the Georgia Institute of Technology campus. The image shows various red-brick buildings, green trees, and a tall smokestack in the distance under a soft, hazy sky. The text is overlaid on this image.

COE 3001

MECHANICS OF DEFORMABLE BODIES

Lecture 18 – Analysis of stress – stress transformation

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Apr. 6, 2026

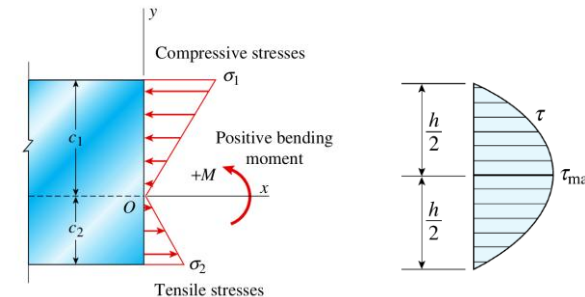
Stress requirements

The normal stress or the shearing stress of a cross section should be smaller than the allowable value.

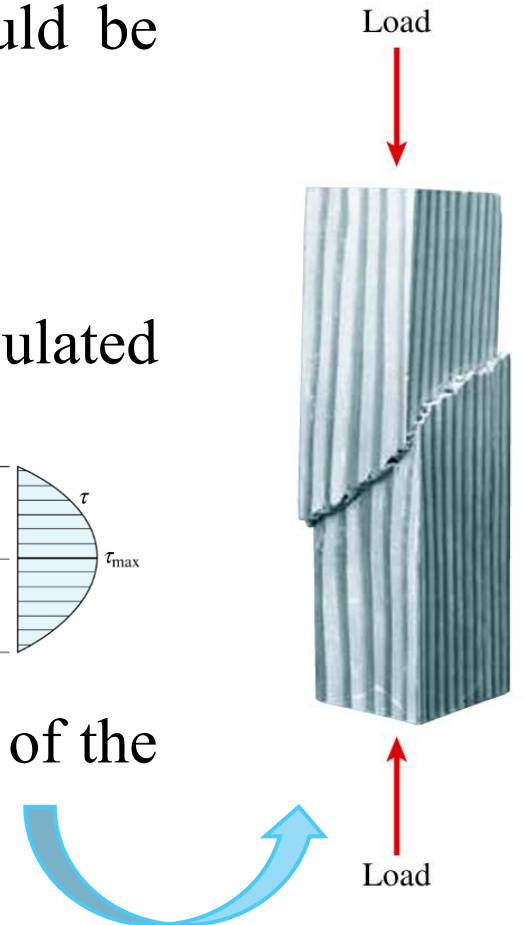
$$|\sigma|_{\max} \leq [\sigma], \quad |\tau|_{\max} \leq [\tau]$$

- Normal and shear stresses in beams, shafts, and bars can be calculated from the basic formulas discussed before, e.g.,

$$\sigma = -\frac{My}{I}, \quad \tau = \frac{VQ}{Ib}$$



- The stresses calculated from these formulas act on cross sections of the members, but larger stresses may occur on inclined sections.



Stress requirements

The normal stress or the shearing stress of a cross section should be smaller than the allowable value.

$$|\sigma|_{\max} \leq [\sigma], \quad |\tau|_{\max} \leq [\tau]$$

- Normal stress from

Therefore, the analysis of stresses and strains begins with a discussion of methods for **finding the normal and shear stresses acting on inclined sections** cut through a member.

$$\sigma = -\frac{My}{I}, \quad \tau = \frac{VQ}{Ib}$$



- The stresses calculated from these formulas act on cross sections of the members, but larger stresses may occur on inclined sections.



Real 3D stress state

The most general state of stress at a point may be represented by 6 components

$$\begin{matrix} \sigma_x, & \sigma_y, & \sigma_z \\ \tau_{xy}, & \tau_{yz}, & \tau_{xz} \end{matrix}$$

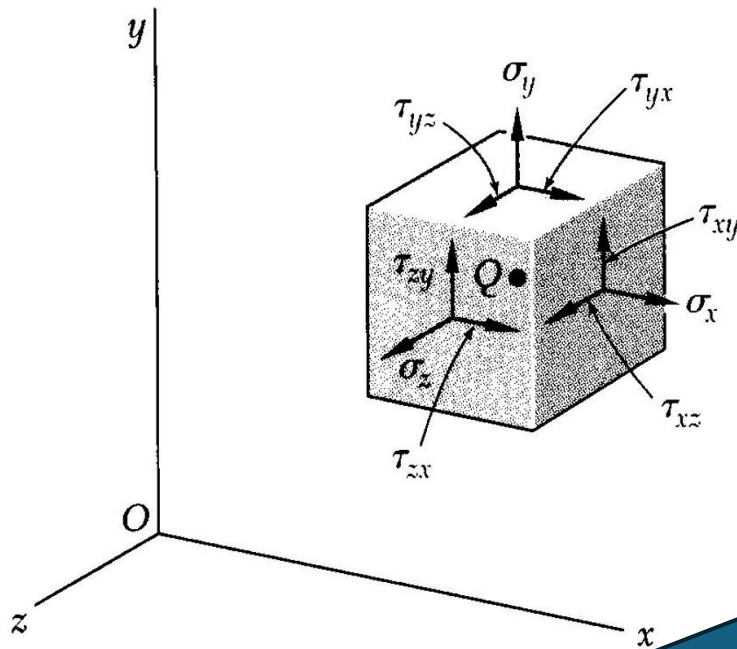
Note:

$$\tau_{xy} = \tau_{yx}, \quad \tau_{yz} = \tau_{zy}, \quad \tau_{xz} = \tau_{zx}$$

τ_{xy} :

- x : plane normal
- y : acting direction

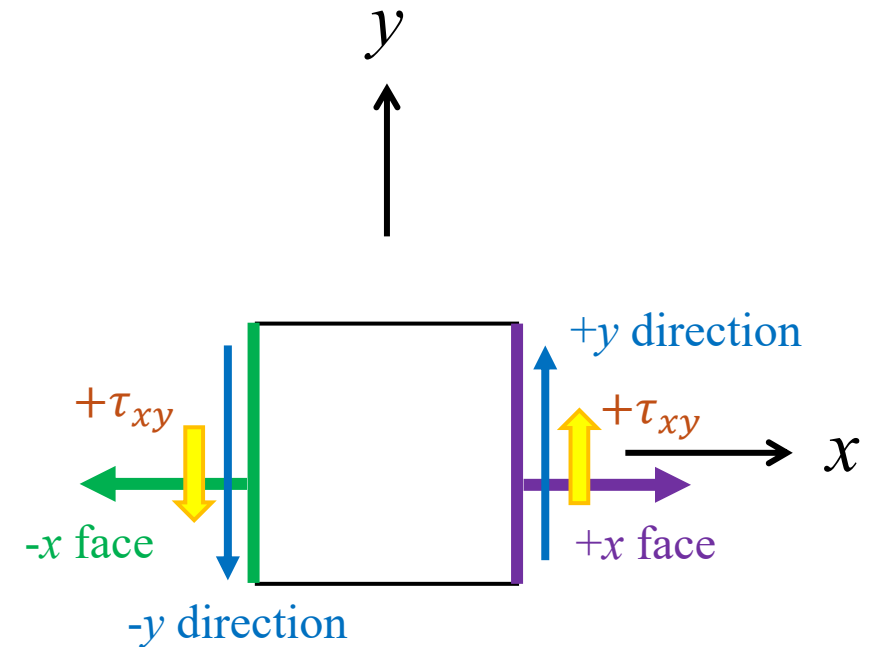
$$\boldsymbol{\sigma} = \begin{bmatrix} \sigma_x & \tau_{xy} & \tau_{xz} \\ \tau_{xy} & \sigma_y & \tau_{yz} \\ \tau_{xz} & \tau_{yz} & \sigma_z \end{bmatrix} \begin{matrix} x \\ y \\ z \end{matrix}$$



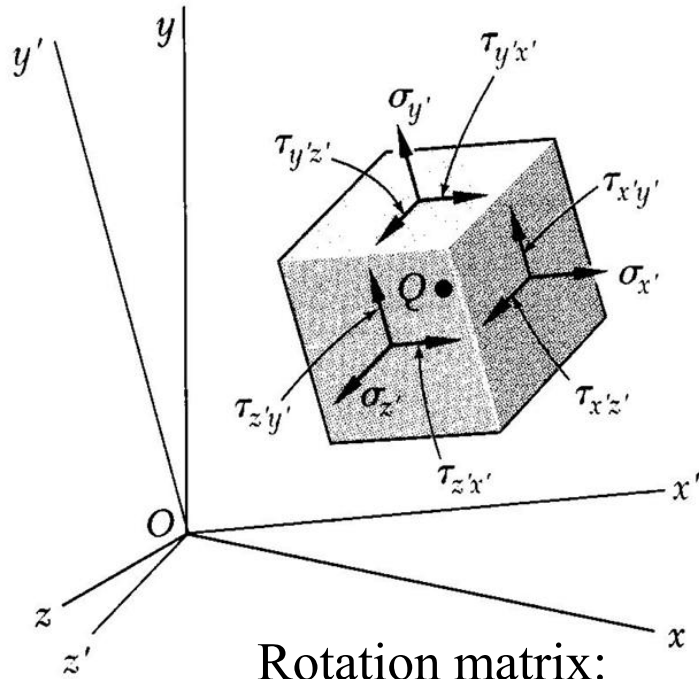
Equality of Shear Stresses
on Perpendicular Planes

Sign convention

- A **normal stress** σ has a subscript that identifies the face on which the stress acts. **Tension is positive and compression is negative**
- A **shear stress** τ has two subscripts, the first subscript denotes the face on which the stress acts, and the second gives the direction on that face.
- A shear stress is **positive when the directions associated with its subscripts are plus-plus or minus-minus**.



3D stress transformation*



Transformation from original coordinates to new coordinates

$$\mathbf{e}'_i = \mathbf{R} \mathbf{e}_i$$

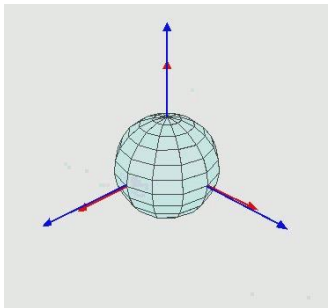
The stress states in two coordinate systems

$$\sigma_{ij} = \mathbf{e}_i^T \boldsymbol{\sigma} \mathbf{e}_j$$

$$\sigma'_{ij} = \mathbf{e}'_i^T \boldsymbol{\sigma}' \mathbf{e}'_j = \mathbf{e}_i^T \mathbf{R}^T \boldsymbol{\sigma}' \mathbf{R} \mathbf{e}_j$$

They describe the same physical stress state

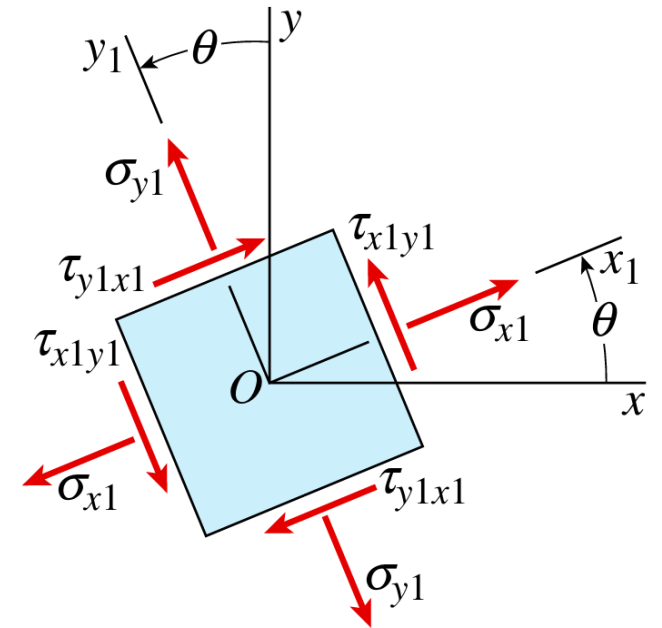
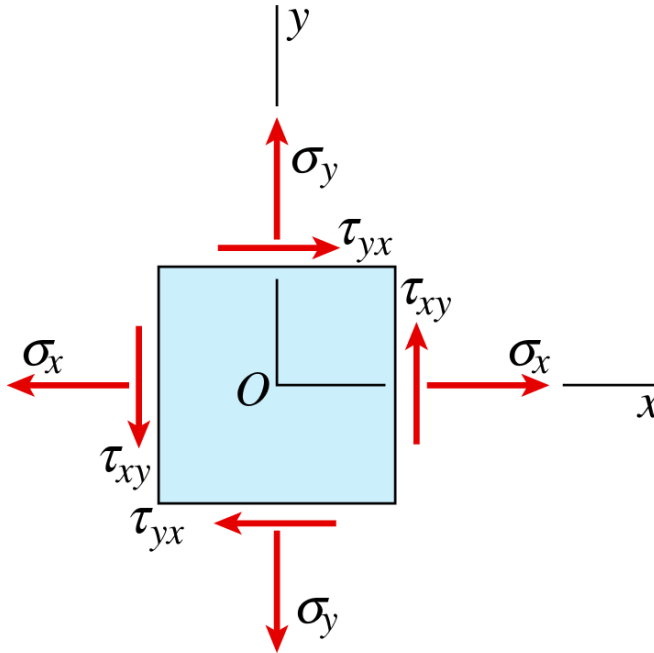
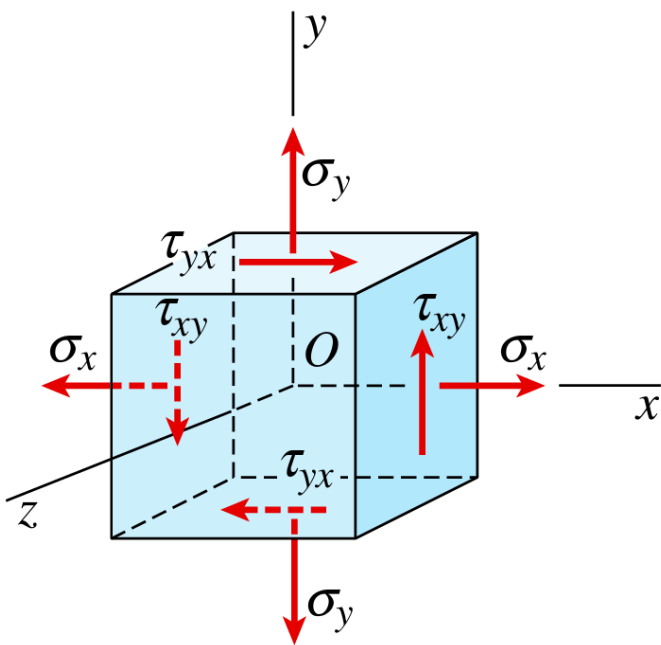
$$\begin{aligned} \mathbf{e}_i^T \mathbf{R}^T \boldsymbol{\sigma}' \mathbf{R} \mathbf{e}_j &= \mathbf{e}_i^T \boldsymbol{\sigma} \mathbf{e}_j \Rightarrow \mathbf{R}^T \boldsymbol{\sigma}' \mathbf{R} = \boldsymbol{\sigma} \\ \Rightarrow \boldsymbol{\sigma}' &= \mathbf{R} \boldsymbol{\sigma} \mathbf{R}^T \end{aligned}$$



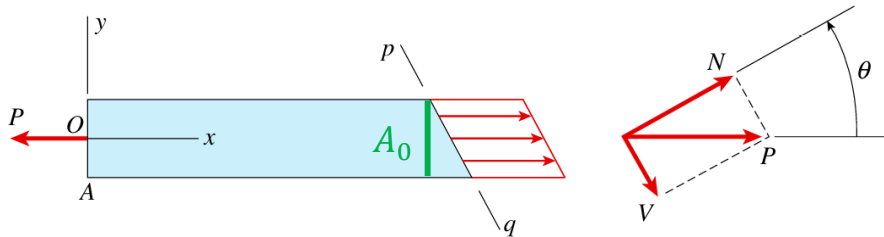
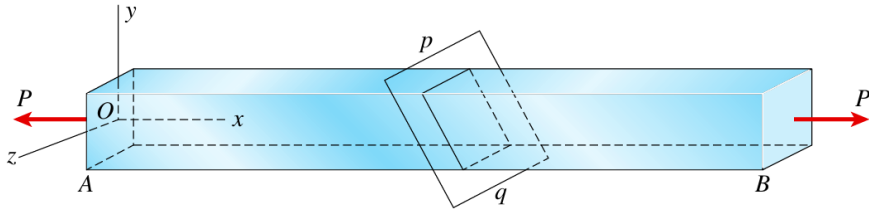
$$\begin{aligned} R &= R_z(\alpha) R_y(\beta) R_x(\gamma) = \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \beta & 0 & \sin \beta \\ 0 & 1 & 0 \\ -\sin \beta & 0 & \cos \beta \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \gamma & -\sin \gamma \\ 0 & \sin \gamma & \cos \gamma \end{bmatrix} \\ &= \begin{bmatrix} \cos \alpha \cos \beta & \cos \alpha \sin \beta \sin \gamma - \sin \alpha \cos \gamma & \cos \alpha \sin \beta \cos \gamma + \sin \alpha \sin \gamma \\ \sin \alpha \cos \beta & \sin \alpha \sin \beta \sin \gamma + \cos \alpha \cos \gamma & \sin \alpha \sin \beta \cos \gamma - \cos \alpha \sin \gamma \\ -\sin \beta & \cos \beta \sin \gamma & \cos \beta \cos \gamma \end{bmatrix} \end{aligned}$$

Plane stress

Plane stress is a state of stress where the stress acting perpendicular to a plane is (assumed to be) zero, simplifying a 3D problem into a 2D one.



Lecture 4 Page 12

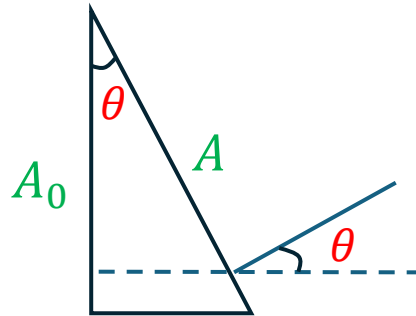


Geometric relation:

$$A = \frac{A_0}{\cos \theta}$$

$$N = P \cos \theta$$

$$V = P \sin \theta$$

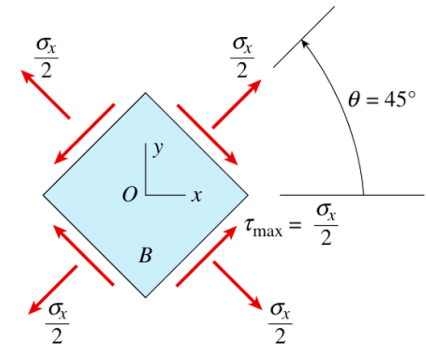
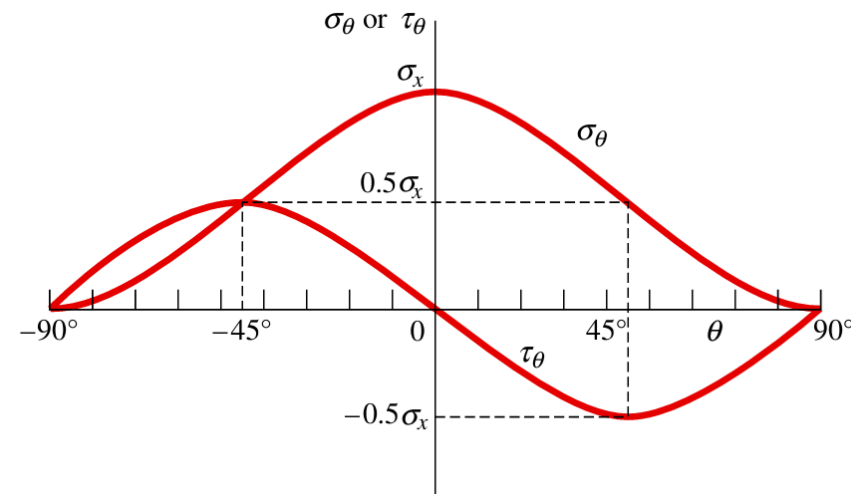


Normal stress:

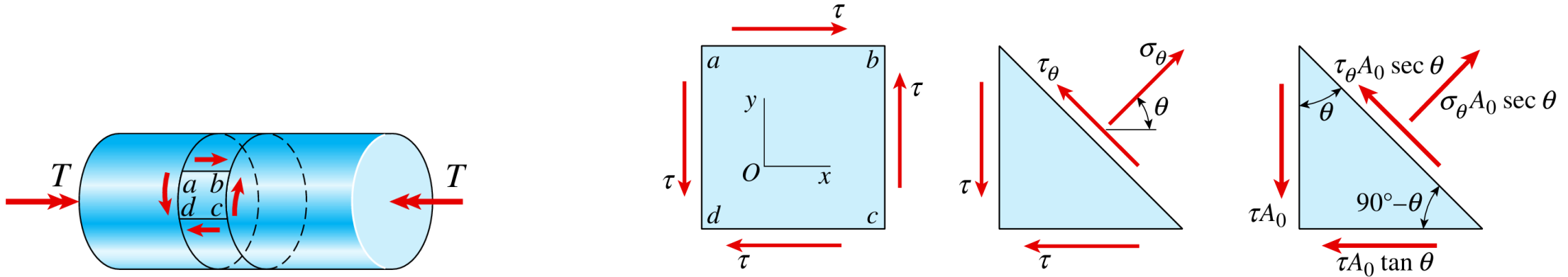
$$\sigma_{\theta} = \frac{N}{A} = \frac{P}{A_0} \cos^2 \theta = \frac{\sigma_x}{2} (1 + \cos 2\theta)$$

Shear stress:

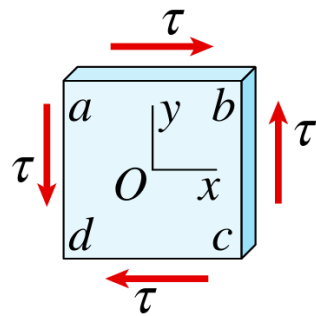
$$\tau_{\theta} = -\frac{V}{A} = -\frac{P}{A} \sin \theta \cos \theta = -\frac{\sigma_x}{2} \sin 2\theta \quad (\text{sign convention})$$



Lecture 9 Page 14



(a)



(b)

$$\sigma_{\theta} A_0 \sec \theta = \tau A_0 \sin \theta + \tau A_0 \tan \theta \cos \theta$$

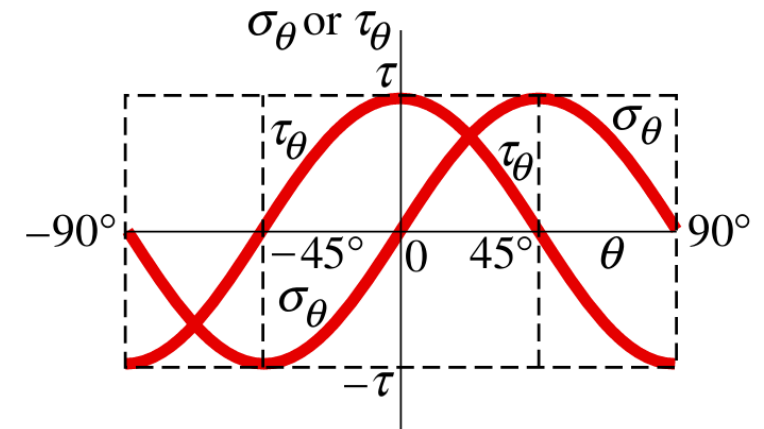
$$\tau_{\theta} A_0 \sec \theta = \tau A_0 \cos \theta - \tau A_0 \tan \theta \sin \theta$$



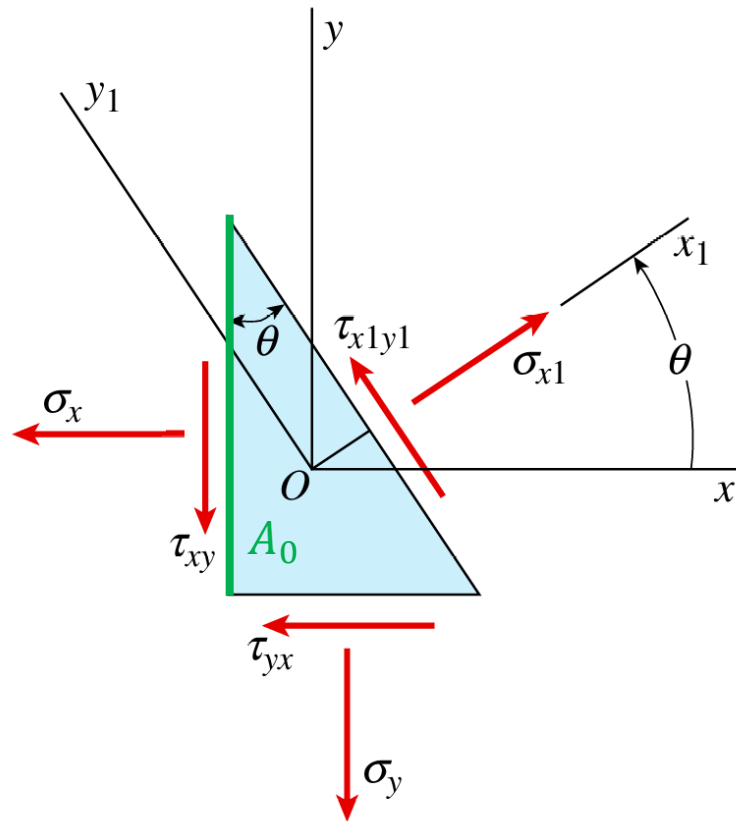
$$\sigma_{\theta} = 2\tau \sin \theta \cos \theta$$

$$\tau_{\theta} = \tau(\cos^2 \theta - \sin^2 \theta)$$

$$\sigma_{\theta} = \tau \sin 2\theta \quad \tau_{\theta} = \tau \cos 2\theta$$



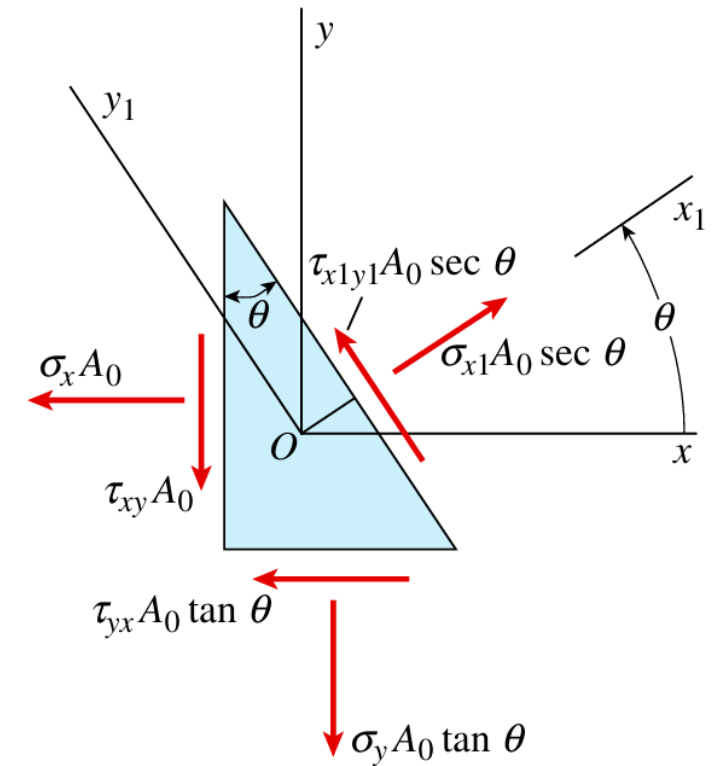
Stresses on inclined sections



Given A_0 , σ_x , σ_y , and τ_{xy}

Find σ_{x1} and τ_{x1y1}

Convert stress into forces



Equilibrium equation in x direction

$$-\sigma_x A_0 - \tau_{xy} A_0 \tan \theta - \frac{\tau_{x1y1} A_0}{\cos \theta} \sin \theta + \frac{\sigma_{x1} A_0}{\cos \theta} \cos \theta = 0$$

Equilibrium equation in y direction

$$-\tau_{xy} A_0 - \sigma_y A_0 \tan \theta + \frac{\tau_{x1y1} A_0}{\cos \theta} \cos \theta + \frac{\sigma_{x1} A_0}{\cos \theta} \sin \theta = 0$$

Solved

$$\sigma_{x1} = \sigma_x \cos^2 \theta + \sigma_y \sin^2 \theta + 2\tau_{xy} \sin \theta \cos \theta$$

$$\tau_{x1y1} = -(\sigma_x - \sigma_y) \sin \theta \cos \theta + \tau_{xy} (\cos^2 \theta - \sin^2 \theta)$$

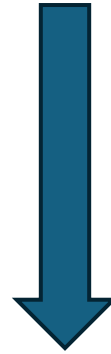
Transformation equations for plane stress

$$\sigma_{x1} = \sigma_x \cos^2 \theta + \sigma_y \sin^2 \theta + 2\tau_{xy} \sin \theta \cos \theta$$

$$\tau_{x1y1} = -(\sigma_x - \sigma_y) \sin \theta \cos \theta + \tau_{xy} (\cos^2 \theta - \sin^2 \theta)$$

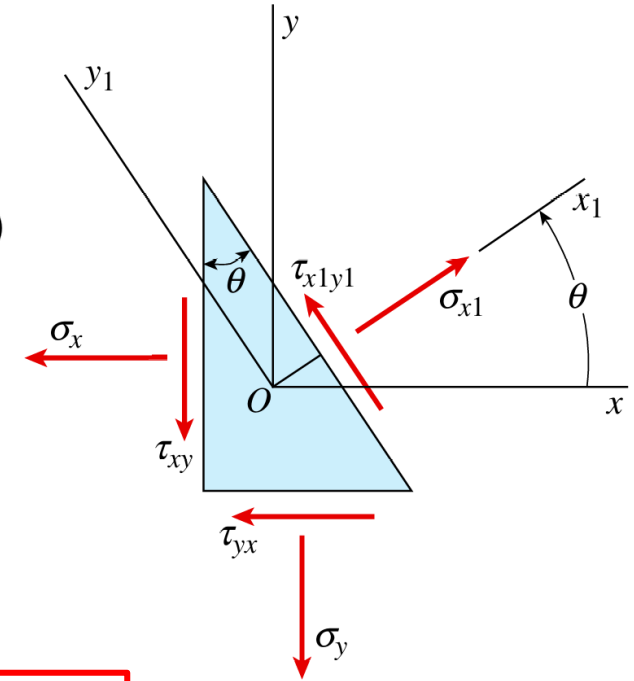
$$\cos^2 \theta = \frac{1}{2}(1 + \cos 2\theta) \quad \sin^2 \theta = \frac{1}{2}(1 - \cos 2\theta)$$

$$\sin \theta \cos \theta = \frac{1}{2} \sin 2\theta$$



$$\sigma_{x1} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos 2\theta + \tau_{xy} \sin 2\theta$$

$$\tau_{x1y1} = -\frac{\sigma_x - \sigma_y}{2} \sin 2\theta + \tau_{xy} \cos 2\theta$$



Mohr's circle

$$\sigma_{x1} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos 2\theta + \tau_{xy} \sin 2\theta$$

$$\tau_{x1y1} = -\frac{\sigma_x - \sigma_y}{2} \sin 2\theta + \tau_{xy} \cos 2\theta$$

$$\sigma_{x1} - \frac{\sigma_x + \sigma_y}{2} = \frac{\sigma_x - \sigma_y}{2} \cos 2\theta + \tau_{xy} \sin 2\theta$$

$$\tau_{x1y1} = -\frac{\sigma_x - \sigma_y}{2} \sin 2\theta + \tau_{xy} \cos 2\theta$$

$$\left(\sigma_{x1} - \frac{\sigma_x + \sigma_y}{2} \right)^2 + (\tau_{x1y1})^2 = \left(\frac{\sigma_x - \sigma_y}{2} \right)^2 + \tau_{xy}^2$$

Center:

$$(\sigma_{\text{aver}}, 0) = \left(\frac{\sigma_x + \sigma_y}{2}, 0 \right)$$

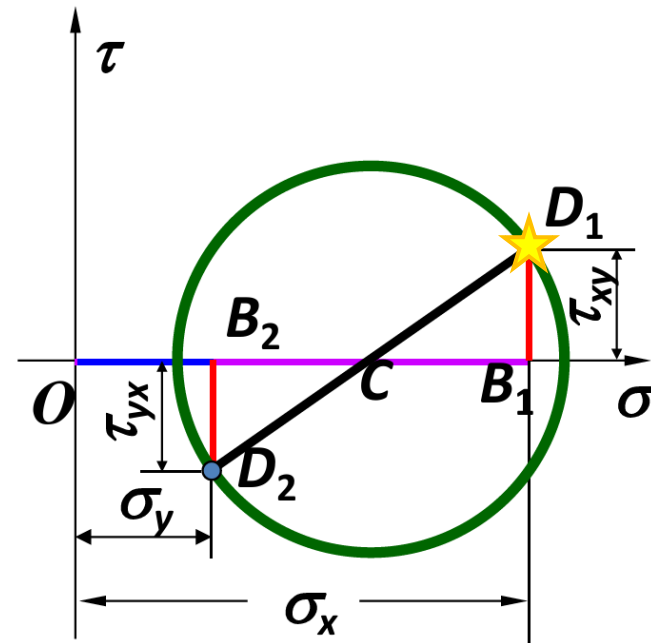
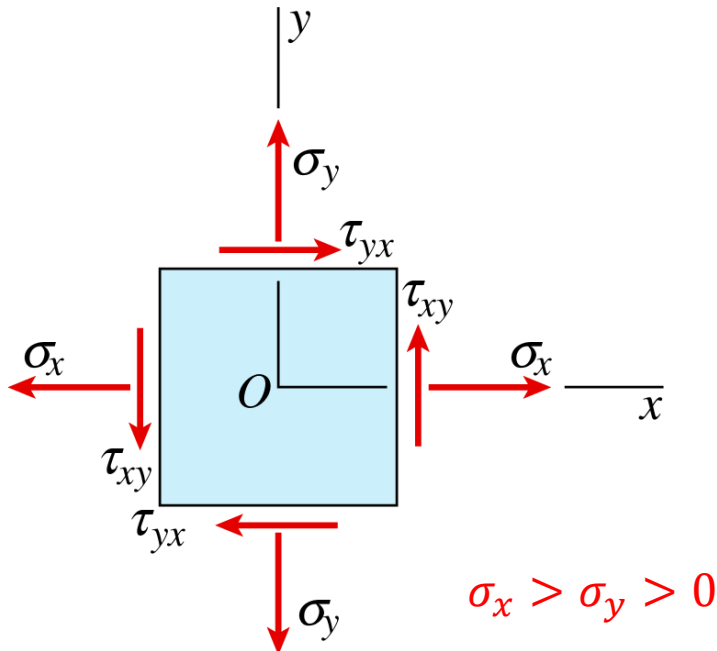
Radius:

$$R = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2} \right)^2 + \tau_{xy}^2}$$

Draw Mohr's circle (1)

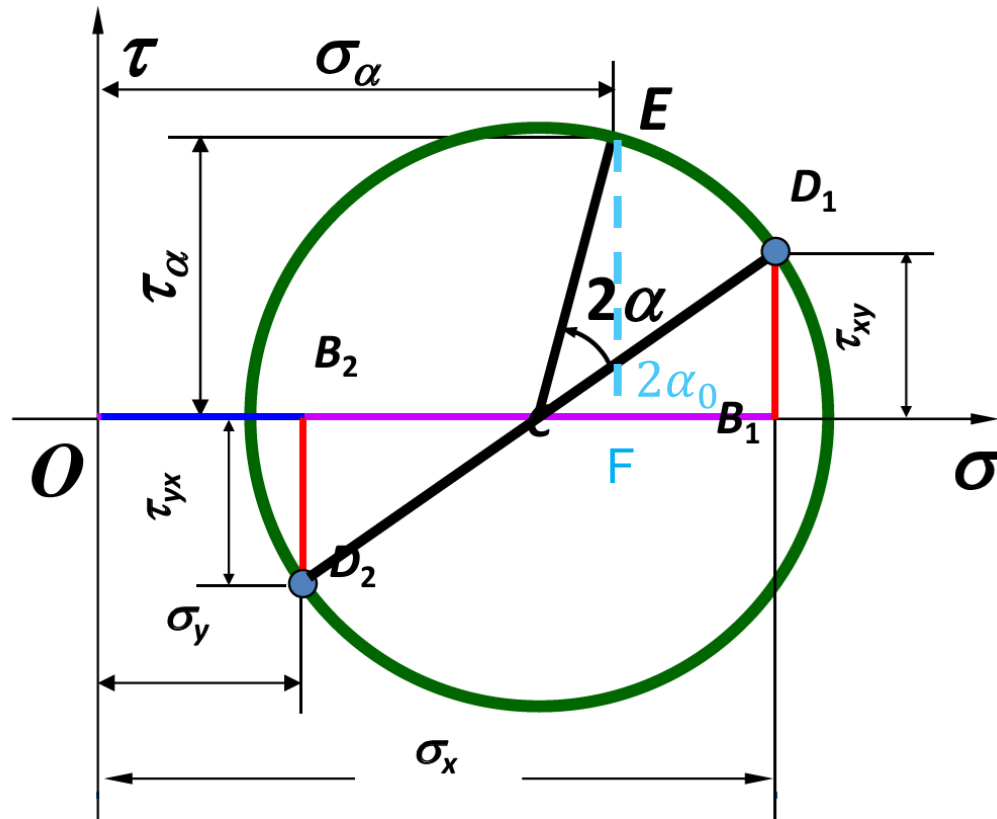
$$(\sigma - \sigma_{\text{aver}})^2 + (\tau)^2 = R^2$$

$$\sigma_{\text{aver}} = \frac{\sigma_x + \sigma_y}{2}, \quad R = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$



The angle correspondence

Coordinate of E



$$\begin{aligned} OF &= OC + CF = OC + CE \cos(2\alpha_0 + 2\alpha) \\ &= OC + CE \cos 2\alpha_0 \cos 2\alpha - CE \sin 2\alpha_0 \sin 2\alpha \\ &= OC + CB_1 \cos 2\alpha - B_1D_1 \sin 2\alpha \end{aligned}$$

$$\Rightarrow \sigma_\alpha = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos 2\alpha - \tau_{xy} \sin 2\alpha$$

$$\begin{aligned} EF &= CE \sin(2\alpha_0 + 2\alpha) \\ &= CE \sin 2\alpha_0 \cos 2\alpha + CE \cos 2\alpha_0 \sin 2\alpha \\ &= B_1D_1 \cos 2\alpha + CB_1 \sin 2\alpha \end{aligned}$$

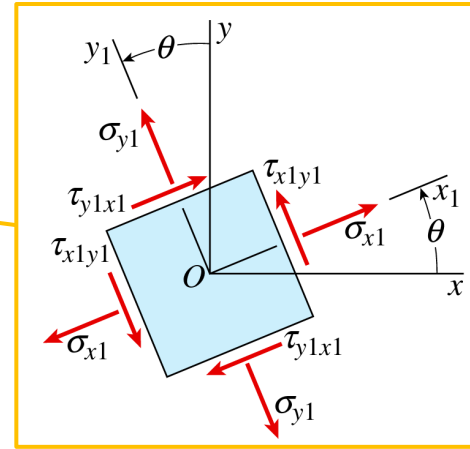
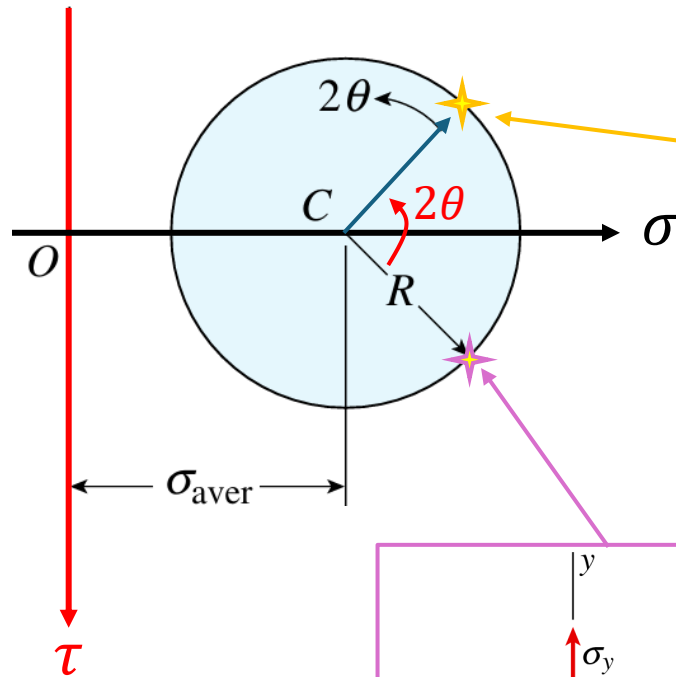
$$\Rightarrow \tau_\alpha = \frac{\sigma_x - \sigma_y}{2} \sin 2\alpha + \tau_{xy} \cos 2\alpha$$

$$\alpha = -\theta$$

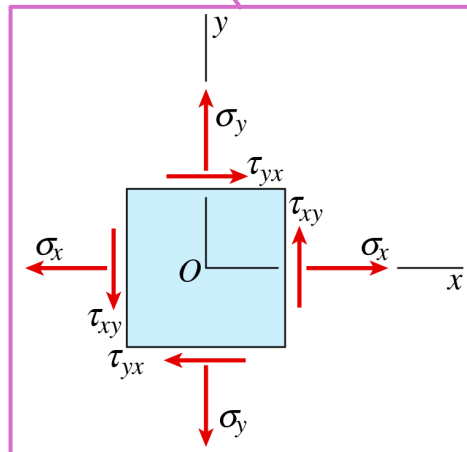
$$\sigma_{x1} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos 2\theta + \tau_{xy} \sin 2\theta$$

$$\tau_{x1y1} = -\frac{\sigma_x - \sigma_y}{2} \sin 2\theta + \tau_{xy} \cos 2\theta$$

Another way to draw Mohr's circle



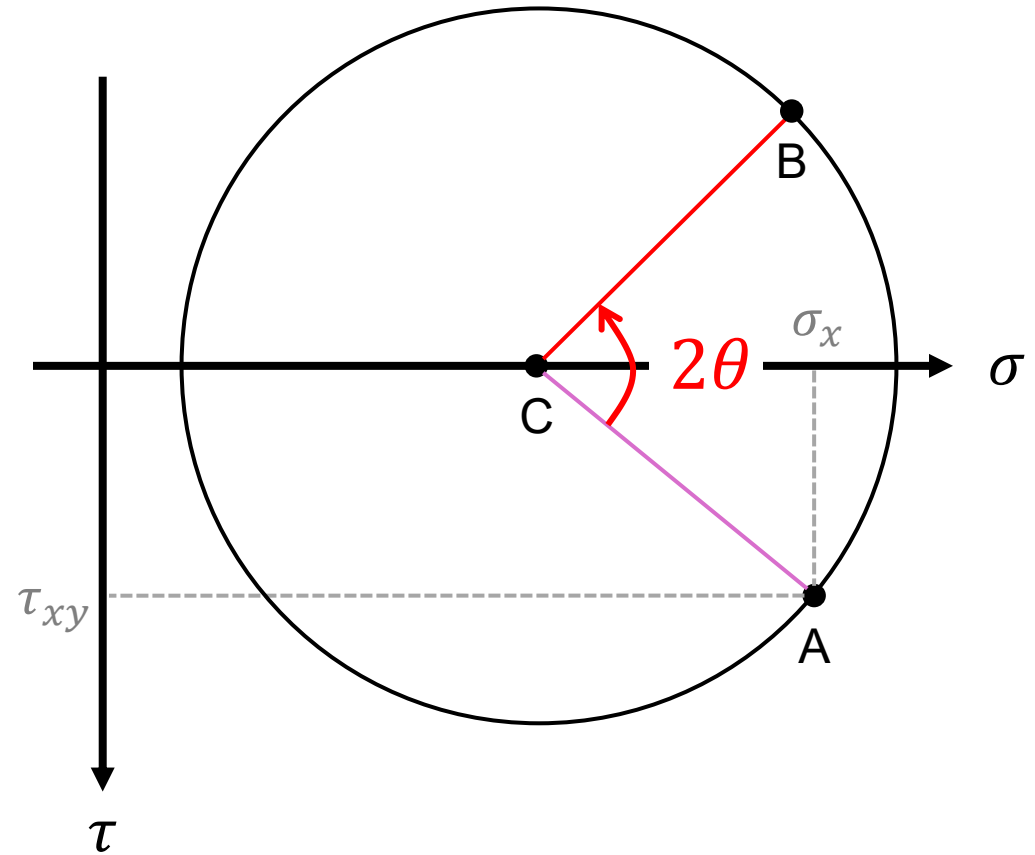
This form of Mohr's circle is used in our course.



The advantage of plotting shear stresses **positive downward** is that the angle 2θ on Mohr's circle will be **positive when counterclockwise**, which agrees with the positive direction of 2θ in the derivation of the transformation equations

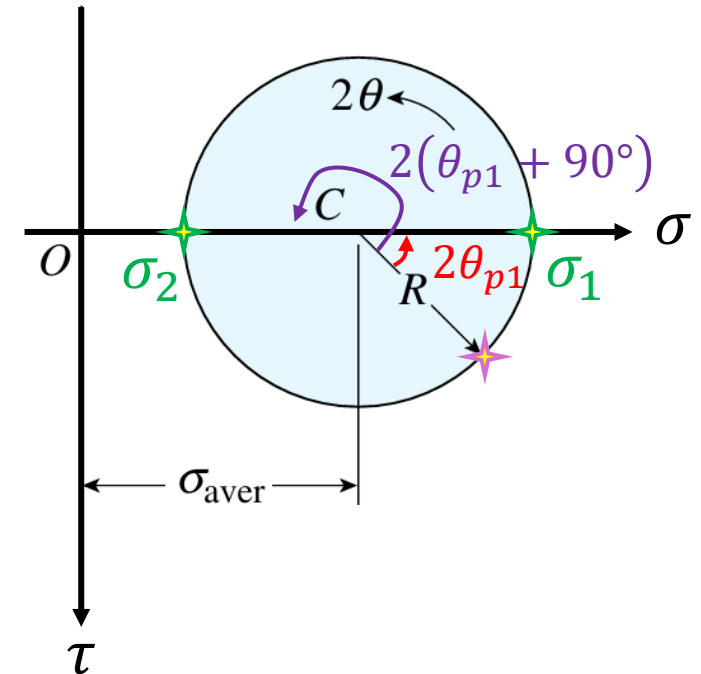
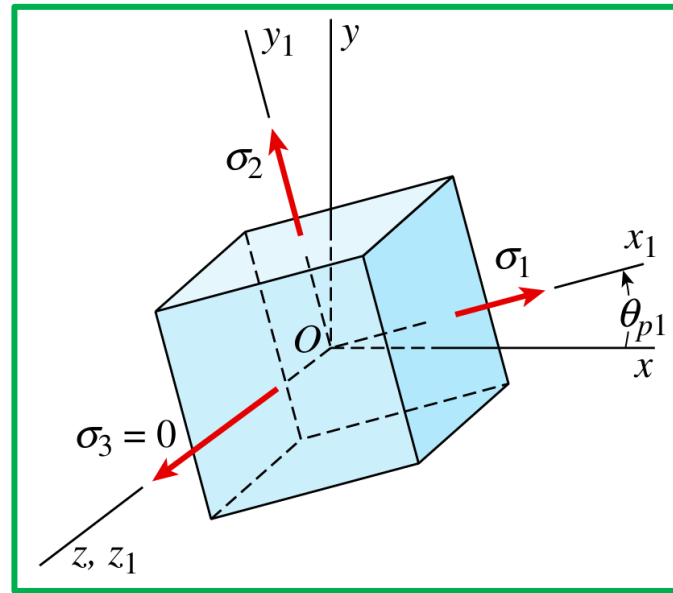
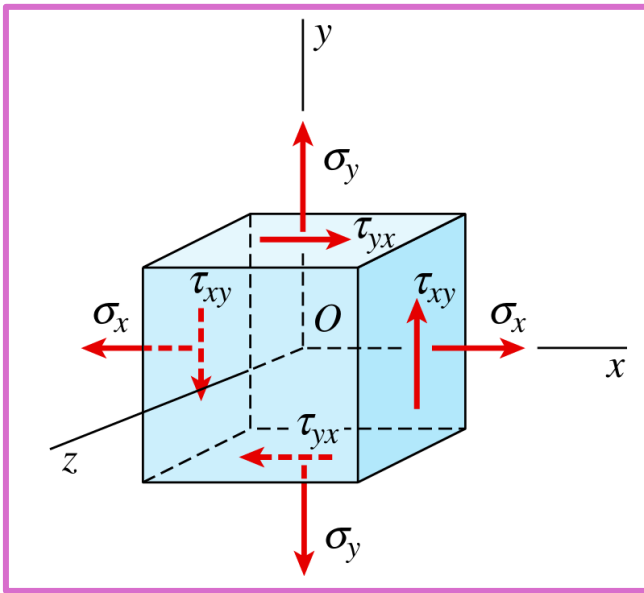
Draw Mohr's circle (2)

- Draw coordinate axes. τ axis positive downward.
- Locate the center C with coordinate $(\sigma_{\text{aver}}, 0)$
- Locate the current stress state point A with coordinate (σ_x, τ_{xy})
- Draw the Mohr's circle with C as center and CA as radius
- Rotate CA by 2θ to get the stress state on inclined plane B

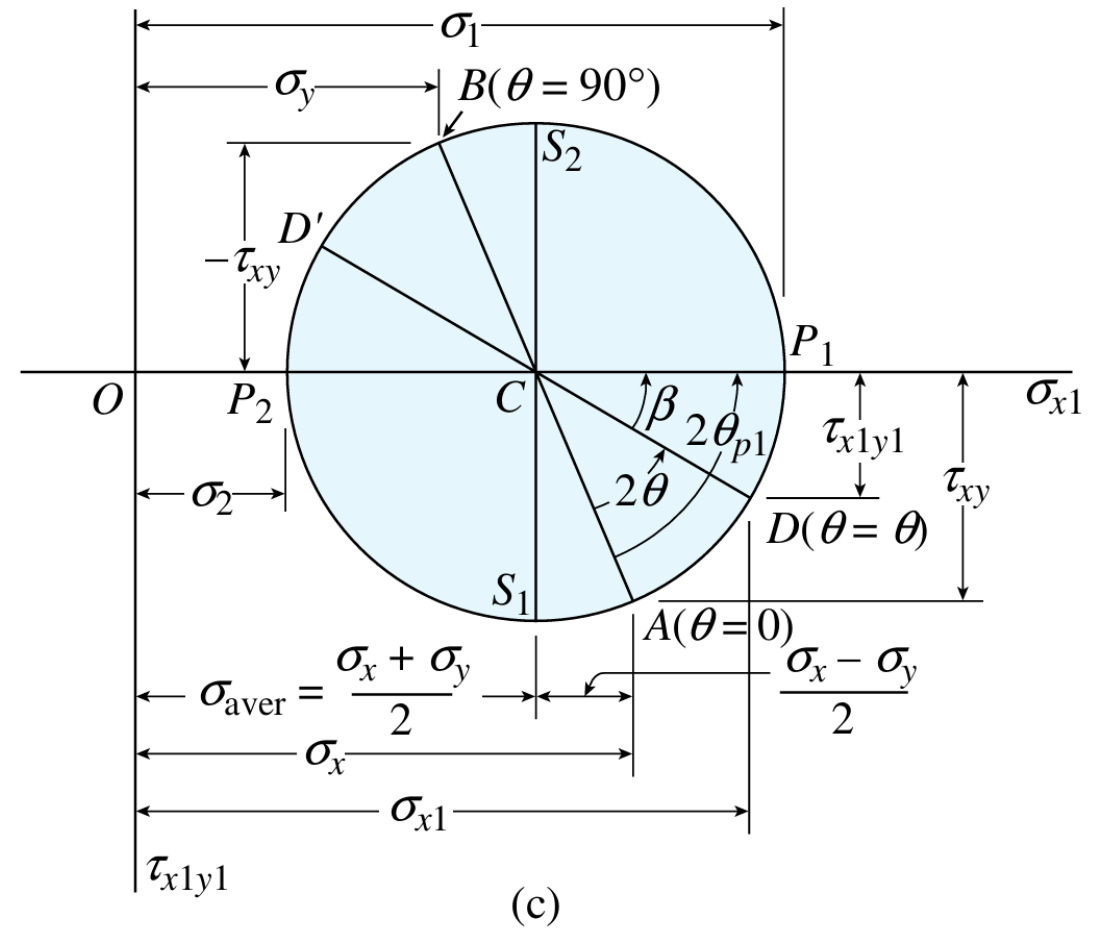
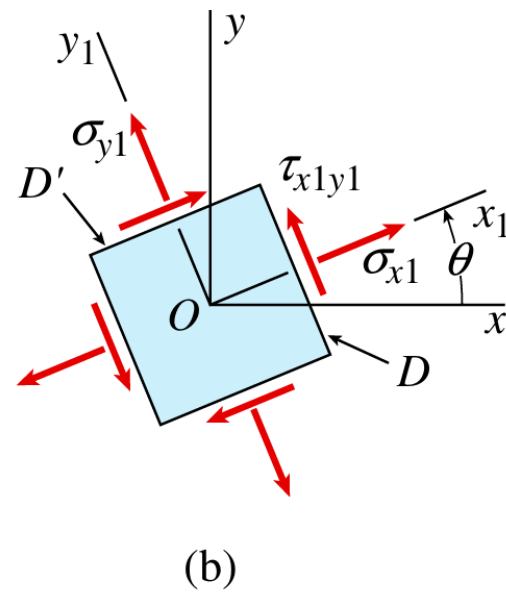
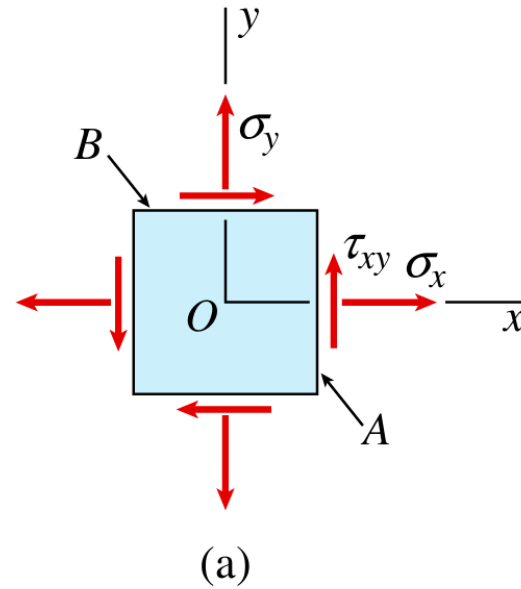


Principal Stresses

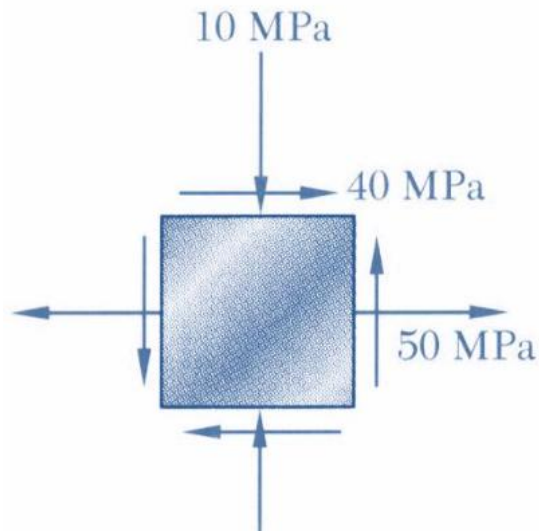
Principal stresses ($\sigma_1, \sigma_2, \sigma_1 > \sigma_2$) are the maximum and minimum normal stresses at an angle, where shear stresses are zero.



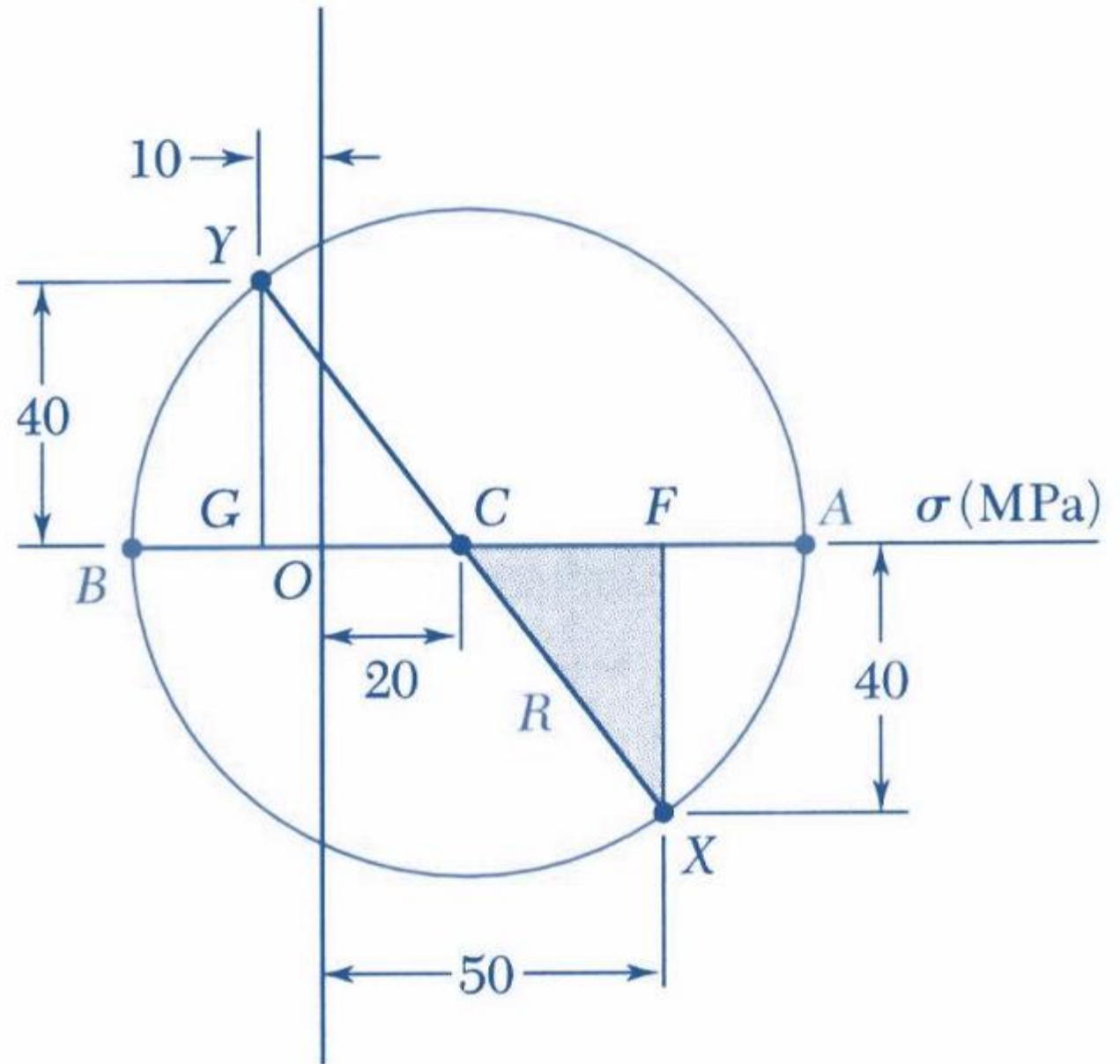
Summary



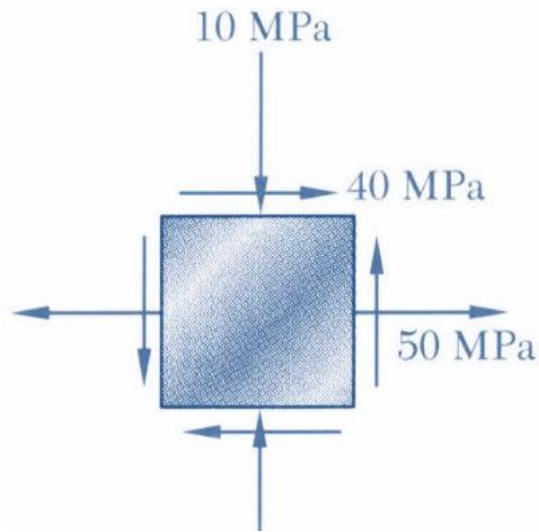
Example



- For the state of plane stress shown,
- (a) construct Mohr's circle,
 - (b) determine the principal planes
 - (c) calculate the principal stresses
 - (d) calculate the maximum shear stress and the corresponding normal stress

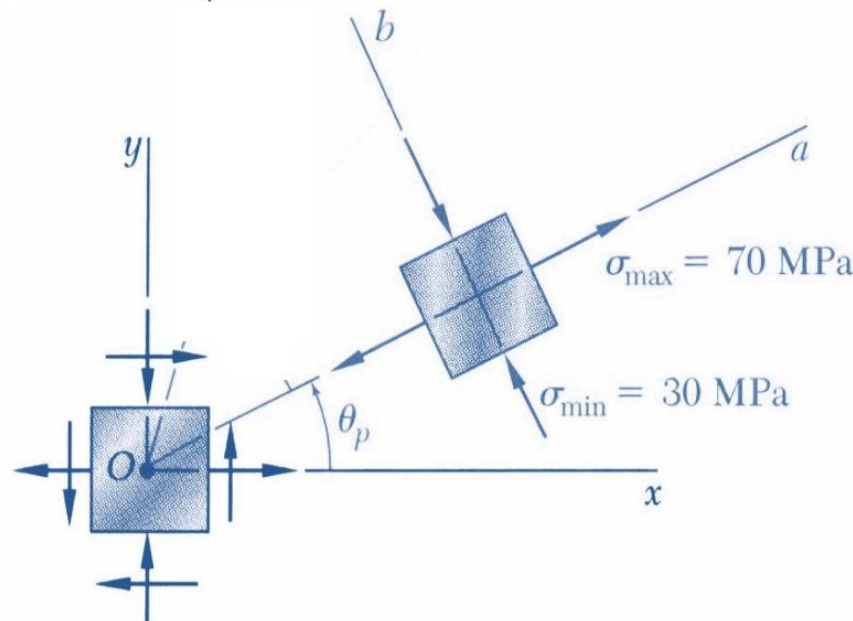
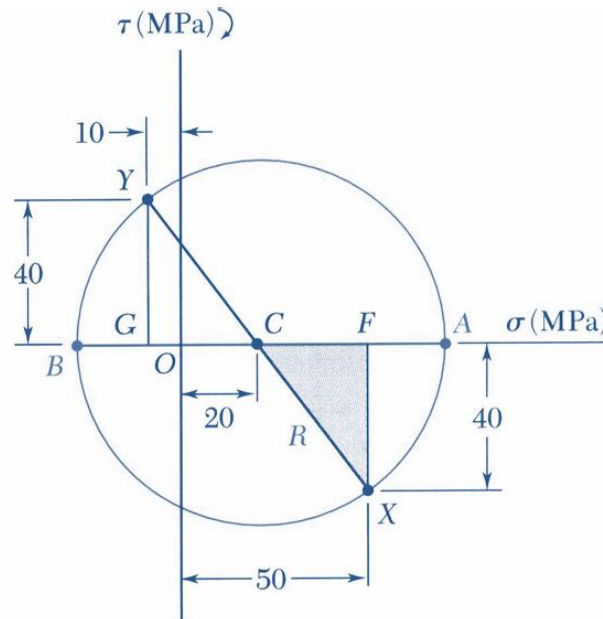


Example



For the state of plane stress shown,

- (a) construct Mohr's circle,
- (b) determine the principal planes
- (c) calculate the principal stresses
- (d) calculate the maximum shear stress and the corresponding normal stress



2) Principal planes and stresses

$$\sigma_{\max} = OA = OC + CA = 20 + 50$$

$$\sigma_{\max} = 70 \text{ MPa}$$

$$\sigma_{\min} = OB = OC - BC = 20 - 50$$

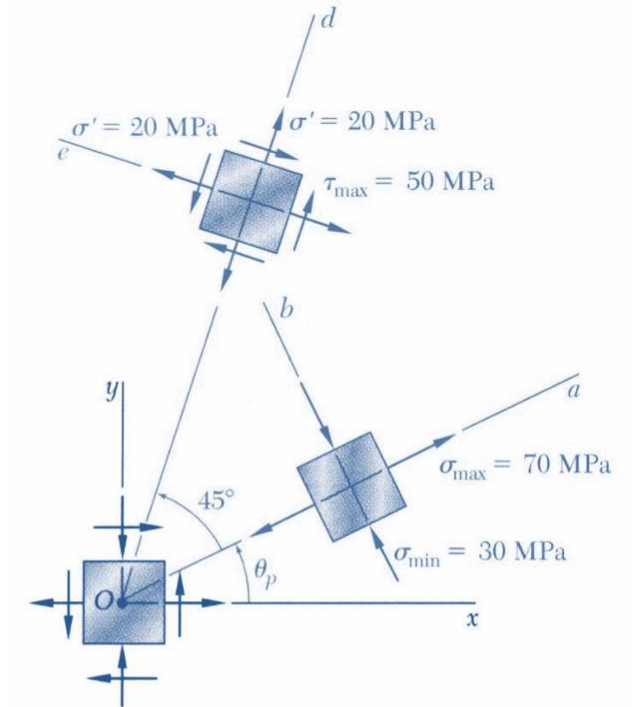
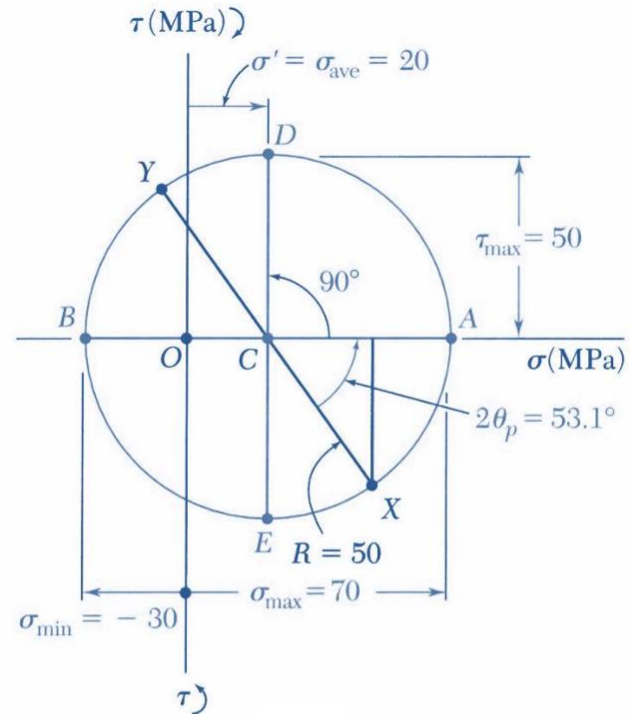
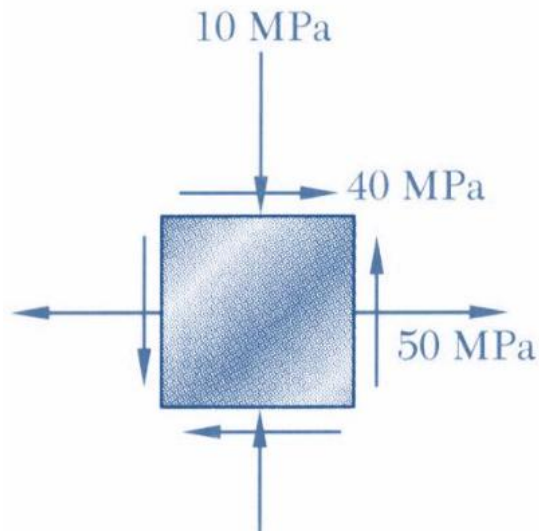
$$\sigma_{\min} = -30 \text{ MPa}$$

$$\tan 2\theta_p = \frac{FX}{CP} = \frac{40}{30}$$

$$2\theta_p = 53.1^\circ$$

$$\theta_p = 26.6^\circ$$

Example



For the state of plane stress shown,

- construct Mohr's circle,
- determine the principal planes
- calculate the principal stresses
- calculate the maximum shear stress and the corresponding normal stress

3) Maximum shear stress

$$\theta_s = \theta_p + 45^\circ$$

$$\theta_s = 71.6^\circ$$

$$\tau_{\max} = R$$

$$\tau_{\max} = 50 \text{ MPa}$$

$$\sigma' = \sigma_{\text{ave}}$$

$$\sigma' = 20 \text{ MPa}$$